

• funkcija stanja $\Psi = \Psi(\vec{r}_1, \dots, \vec{r}_n)$

• Schrödinger it $\frac{\partial \Psi}{\partial t} = H \Psi$

• neodvisni delci - faktorizirano Ψ

$$\Psi(\vec{r}_1, \dots, \vec{r}_N) = \Psi_\alpha(\vec{r}_1) \times \Psi_\beta(\vec{r}_2) \times \dots \times \Psi_\gamma(\vec{r}_N)$$

• fermioni: $e^-, p, n, \dots$ polcel spin($\hbar/2$) **PAULIJEVA PREPOVED**

• bozoni: foton, ${}^4\text{He}, \dots$ cel spin($\hbar$) **NI PAULIJEVE PREPOVEDI**

• $\Psi(\vec{r}_1, \vec{r}_2) \propto \Psi_\alpha(\vec{r}_1) \Psi_\beta(\vec{r}_2) + \Psi_\beta(\vec{r}_1) \Psi_\alpha(\vec{r}_2)$ primerno simetrizirana funkc. stanja za bozone:
invariantna na zamenjavo $1 \leftrightarrow 2$

• $\Psi(\vec{r}_1, \vec{r}_2) \propto \Psi_\alpha(\vec{r}_1) \Psi_\beta(\vec{r}_2) - \Psi_\beta(\vec{r}_1) \Psi_\alpha(\vec{r}_2)$ - fermione: spremenjeni znak pri zamenjavi $1 \leftrightarrow 2$

• $\Psi(\vec{r}_1, \vec{r}_2, \vec{r}_3) \propto \Psi_\alpha(\vec{r}_1) \Psi_\beta(\vec{r}_2) \Psi_\gamma(\vec{r}_3) + \Psi_\beta(\vec{r}_1) \Psi_\alpha(\vec{r}_2) \Psi_\gamma(\vec{r}_3) + \Psi_\gamma(\vec{r}_1) \Psi_\beta(\vec{r}_2) \Psi_\alpha(\vec{r}_3)$

$+ \Psi_\alpha(\vec{r}_1) \Psi_\gamma(\vec{r}_2) \Psi_\beta(\vec{r}_3) + \Psi_\gamma(\vec{r}_1) \Psi_\alpha(\vec{r}_2) \Psi_\beta(\vec{r}_3) + \Psi_\beta(\vec{r}_1) \Psi_\gamma(\vec{r}_2) \Psi_\alpha(\vec{r}_3)$

--- - za bozone

--- - za fermione

$\rightarrow$ $N!$ produktnih funkc. stanja

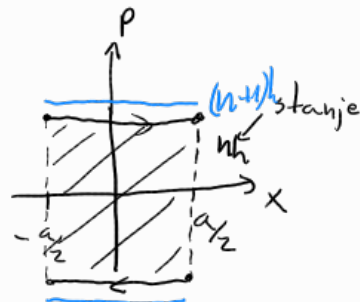
Bohr-Sommerfeldovo kvantizacijsko pravilo

$$\oint p dx = nh, \quad n=0, 1, 2, \dots$$

$\oint$

1 obhod/cikel
pri vezanem stanju

• $\propto$ pot. jama

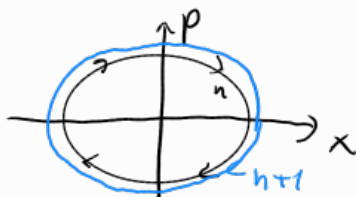


$$2p_n a = nh$$

$$\rightarrow \boxed{E_n = \frac{n^2 h^2}{8ma^2}}$$

$\rightarrow$ 1 stanje zavzema ploščino
 h v faznem prostoru

• harmonični oscilator



$$E_n = n\hbar\omega \leftarrow \text{BS pravilo}$$

$$E_n = (n + \frac{1}{2})\hbar\omega \leftarrow \text{točen rezultat}$$

1 delec, 1D: h

1 delec, 3D: h^3

N delcev, 3D: h^{3N}

• fazni integral

$$e^{-\beta F} = C \int \exp(-\beta E) d\Gamma = \sum_i \exp(-\beta E_i)$$

po vseh
stanjih

$$C = \frac{(2j+1)^N}{N! h^{3N}} \leftarrow \text{spinska degeneracija}$$

simetrizacija večdelnih
funkcij stanja

volumen 1 stanja (kvantizacija faznega prostora)

$$\langle y \rangle = \sum_i y_i \exp(-\beta E_i)$$

Dvoatomni idealni plin

vibracije
rotacija
translacija

$$e^{-\beta F} = \left[\int e^{-\beta E_{tr}} d\Gamma_{tr} \right] \times e^{-\beta F_{rot}} \times e^{-\beta F_{vib}}$$

• Translacija

• ∞ pot. jama

$$E_n = \frac{h^2 k^2}{8ma^2} \leftarrow \text{značilna energija}$$

$\ll$

$\Rightarrow$ klasičen opis bo ok

$$e^{-\beta F_{tr}} = C V \left(\frac{2\pi m}{\beta} \right)^{3/2}$$

$$\beta F_{tr} = -\ln C - \ln V + \frac{3}{2} \ln \beta - \frac{3}{2} \ln 2\pi m$$

$$\bar{E}_4 = \frac{3}{2} \beta^{-1} = \frac{3}{2} k_B T \rightarrow C_v^{tr} = \frac{3}{2} k_B$$

• Rotacija

$$E_j = \frac{j(j+1)\hbar^2}{2I}$$

vrtilna
količina, $(2j+1)$ projekcij
vztrajnostni moment

$$e^{-\beta F_{\text{rot}}} = \sum_j \sum_{m_j} \exp\left(-\beta \frac{j(j+1)\hbar^2}{2I}\right) = \sum_j (2j+1) \exp\left(-\beta \frac{j(j+1)\hbar^2}{2I}\right) \stackrel{1/\beta_{\text{rot}}}{\sim}$$

$$k_B T_{\text{rot}} = \frac{1}{\beta_{\text{rot}}} = \frac{\hbar^2}{2I}$$

• $\beta \ll \beta_{\text{rot}}$:

$$e^{-\beta F_{\text{rot}}} \approx \int_0^\infty (2j+1) \exp(-j(j+1) \beta/\beta_{\text{rot}}) dj =$$

$$= \int_0^\infty \exp(-\beta/\beta_{\text{rot}} u) du =$$

$$= \frac{\beta_{\text{rot}}}{\beta} \rightarrow \beta F_{\text{rot}} = \ln \beta - \ln \beta_{\text{rot}}$$

$$\bar{E}_{\text{rot}} = \beta^{-1} = k_B T$$

$$C_v^{\text{rot}} = k_B$$

• $\beta \gg \beta_{\text{rot}}$:

$$e^{-\beta F_{\text{rot}}} \approx 1 + 3 \exp(-2\beta/\beta_{\text{rot}}) + \dots$$

$$\beta F_{\text{rot}} \approx -\ln(1 + 3 \exp(-2\beta/\beta_{\text{rot}}))$$

$$\bar{E}_{\text{rot}} \approx -\frac{3(-2) \frac{1}{\beta_{\text{rot}}} \exp(-2\beta/\beta_{\text{rot}})}{1 + 3 \exp(-2\beta/\beta_{\text{rot}})} \approx 6 k_B T_{\text{rot}} \exp(-2 \frac{T_{\text{rot}}}{T})$$

$$C_v^{\text{rot}} = 12 \left(\frac{T_{\text{rot}}}{T}\right)^2 k_B \exp(-2 \frac{T_{\text{rot}}}{T})$$

• Vibracije

• harmonični oscilator

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

$$e^{-\beta F_{\text{vib}}} = \sum_n \exp(-(n + \frac{1}{2}) \beta \hbar \omega) = \frac{1}{\beta_{\text{vib}}} = k_B T_{\text{vib}} = \hbar \omega$$

$$= e^{-\frac{1}{2} \beta/\beta_{\text{vib}}} \underbrace{\sum_{n=0}^{\infty} e^{-n \beta/\beta_{\text{vib}}}}_{\text{geom. vrsta}} = e^{-\frac{1}{2} \beta/\beta_{\text{vib}}} \cdot \frac{1}{1 - e^{-\beta/\beta_{\text{vib}}}}$$

$$\beta F_{\text{vib}} = \frac{1}{2} \frac{\beta}{\beta_{\text{vib}}} + \ln(1 - e^{-\beta/\beta_{\text{vib}}})$$

$$\bar{E}_{\text{vib}} = \frac{1}{2\beta_{\text{vib}}} + \frac{\frac{1}{\beta_{\text{vib}}} e^{-\beta/\beta_{\text{vib}}}}{1 - e^{-\beta/\beta_{\text{vib}}}} = \frac{k_B T_{\text{vib}}}{2} + k_B T_{\text{vib}} \frac{1}{e^{T_{\text{vib}}/T} - 1}$$

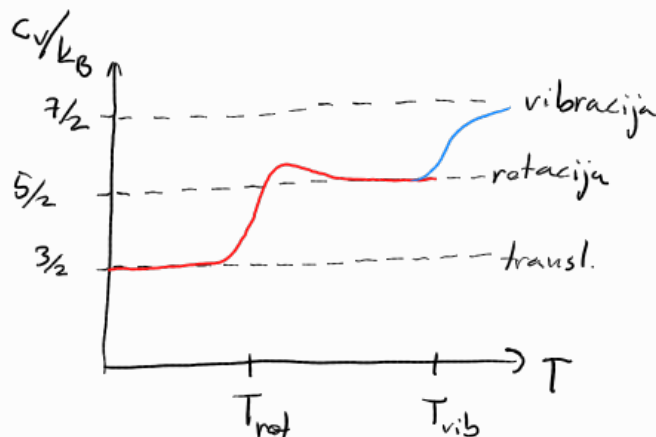
$$C_v^{\text{vib}} = k_B \left(\frac{T_{\text{vib}}}{T}\right)^2 \frac{e^{T_{\text{vib}}/T}}{(e^{T_{\text{vib}}/T} - 1)^2}$$

• $T \ll T_{\text{vib}}$:

$$C_v^{\text{vib}} \approx k_B \left(\frac{T_{\text{vib}}}{T}\right)^2 e^{-T_{\text{vib}}/T}$$

• $T \gg T_{\text{vib}}$:

$$C_v^{\text{vib}} \approx k_B \left(\frac{T_{\text{vib}}}{T}\right)^2 \frac{1 + \frac{T_{\text{vib}}}{T}}{\left(\frac{T_{\text{vib}}}{T}\right)^2} = k_B$$



za H_2 :

$$T_{\text{rot}} = \frac{\hbar^2}{2Ik_B} \lesssim 80\text{K}$$

$$T_{\text{vib}} \sim 1000\text{K}$$

→ navadno smo v režimu $C_v = \frac{5}{2} k_B$

→ $T < T_{\text{rot}}$ - navadno plin

→ $T > T_{\text{vib}}$ - molekula pogosto
kondenzira
že razpade